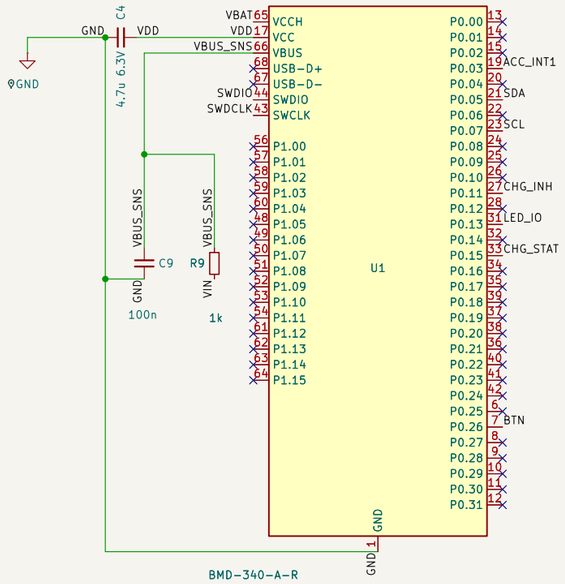
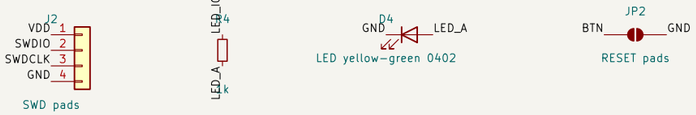
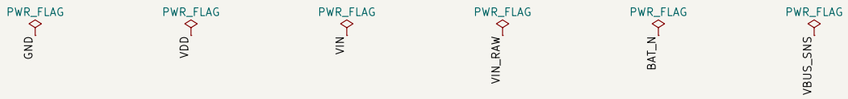
MCU / radio (nRF52840 high-voltage mode)



Programming & status LED



Power flags (ERC)

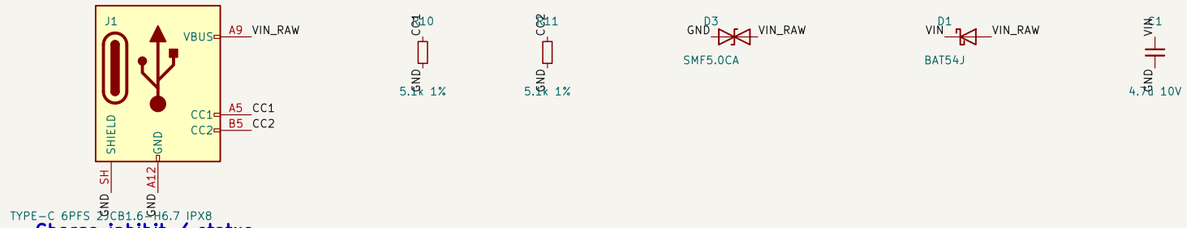


design.py — ShotPuck circuit, single source of truth.

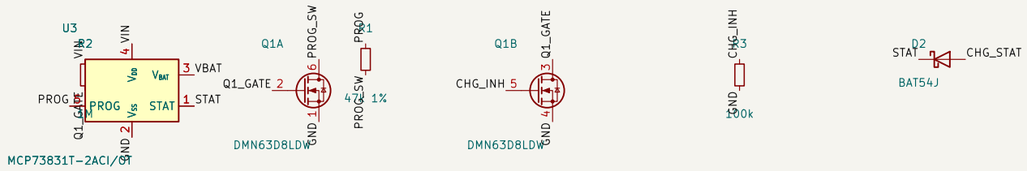
gen_sch.py turns this into shotpuck.kicad_sch and gen_pcb.py into
shotpuck.kicad_pcb. Change the circuit here, then re-run both.

Power path
  J1 USB-C power-only receptacle (VBUS = VIN_RAW) on the left rim, plug through the cap wall:
  R10/R11 5.1k pull CC1/CC2 down so USB-C sources turn on 5 V. D3 SMF5.0CA bidirectional TVS
  to GND (ESD, hot-plug) -> D1 BAT54J (blocks back-feed into the port) -> VIN
  VIN -> R9 1k -> VBUS_SNS (C9 100n) -> U1 VBUS: charger detect only (USB unused, ~24 uA),
  the RC keeps hot-plug spikes off the 5.8 V abs-max VBUS pin
  VIN -> U3 MCP73831-2 (4.20 V, 21 mA via R1 = 47k) -> VBAT (LIR1254, 45-65 mAh)
  VBAT -> U1 VCCH (u-blox BMD-340, nRF52840 high-voltage mode, REG0 -> VCC = VDD 3.0 V)
  VDD -> U2 LSM6DSV32X (VDD + VDDIO), LED, SWD VCC sense
IMU
  U2 LSM6DSV32X on I2C at 0x6A (SA0 = GND, CS = VDD selects I2C). Aux master
  pins SDx/SCx to GND; INT2, OCS_Aux, SDO_Aux unconnected.
Charge inhibit (temperature, firmware)
  PROG -> R1 -> Q1A drain; Q1A gate pulled to VIN by R2 (charging allowed
  whenever a cable is present). Q1B (gate = CHG_INH, R3 pull-down) pulls
  Q1A gate low -> PROG floats -> MCP73831 charge disabled.
Cell protection (PCM: LIR cells have no protection of their own)
  U4 BQ29700 + Q2 DMN2004DWK in the cell's negative lead: BAT_N -> Q2B (DOUT,
  discharge) -> PCM_D -> Q2A (COUT, charge) -> GND. OVP 4.275 V, UVP 2.80 V,
  OCD/OCC +-100 mV across ~1 ohm of FETs (~0.1 A), SCD 0.5 V. R7 330R + C8
  100n filter the BAT supply, R8 2k2 on V-. Standby 4 uA.
Current measurement
  JP1 (bridged solder jumper) in series with the cell: cut, insert an
  ammeter, re-bridge with solder.
Charge status
  STAT -> D2 (cathode) ; D2 anode -> CHG_STAT GPIO with pull-up enabled only
  while VBUS is present (no standby leakage).

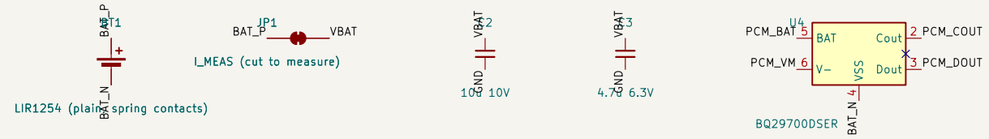
Charging input & charger



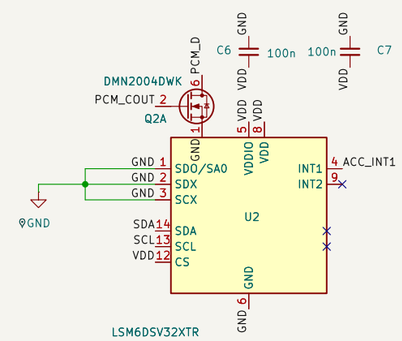
Charge inhibit / status

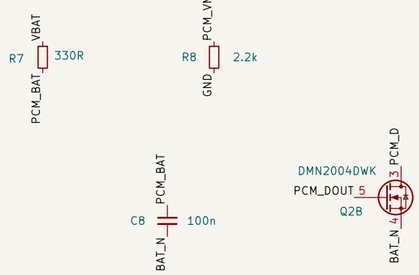


Battery + protection (PCM)

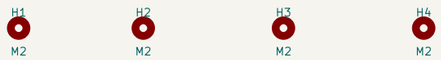


IMU (accelerometer + gyroscope)





Mechanical



Generated by hardware/gen_sch.py from design.py

Sheet: /
File: shotpuck.kicad_sch

Title: ShotPuck — arrow shot counter

Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.6		Id: 1/1